

123456

A

B

C

D

Power Section

Power

File: Power.kicad_sch

GND&NC

File: GND&NC.kicad_sch

User_IO/Banks

USER_IO_Bank_T

File: USER_IO_Bank_B.kicad_sch

USER_IO_Bank_R

File: USER_IO_Bank_L.kicad_sch

USER_IO_Bank_L

File: USER_IO_Bank_R.kicad_sch

USER_IO_Bank_B

File: USER_IO_Bank_T.kicad_sch

Device_Managemant/Configuration

USER_Config

File: USER_Config.kicad_sch

Sheet: /
File: MACHX02_test.kicad_sch

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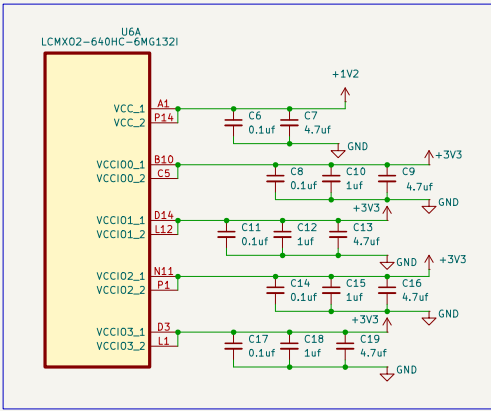
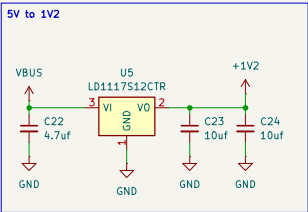
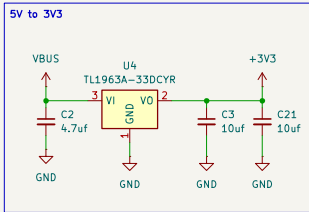
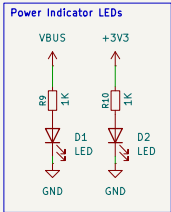
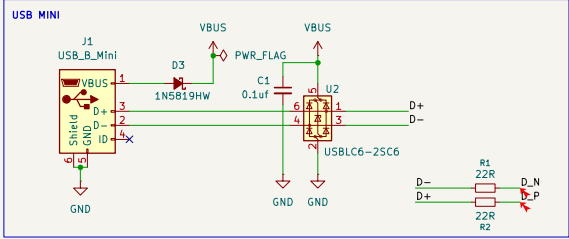
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Id: 1/8

6



Bank	Typical devices connected	Recommended VCCIO Net
PB* (left)	3.3 V sensors, headers, SPI flash (if 3.3 V)	3.3 V
PL* (lower)	Low-voltage memory or ADCs	1.8 V (if memory requires 1.8 V)
PR* (right)	High-speed interfaces – check signalling standard	1.8 V or 2.5 V as required
PT* (top)	UART, SPI, UART–USB level translators	3.3 V

Sheet: /Power/
File: Power.kicad_sch

Title:

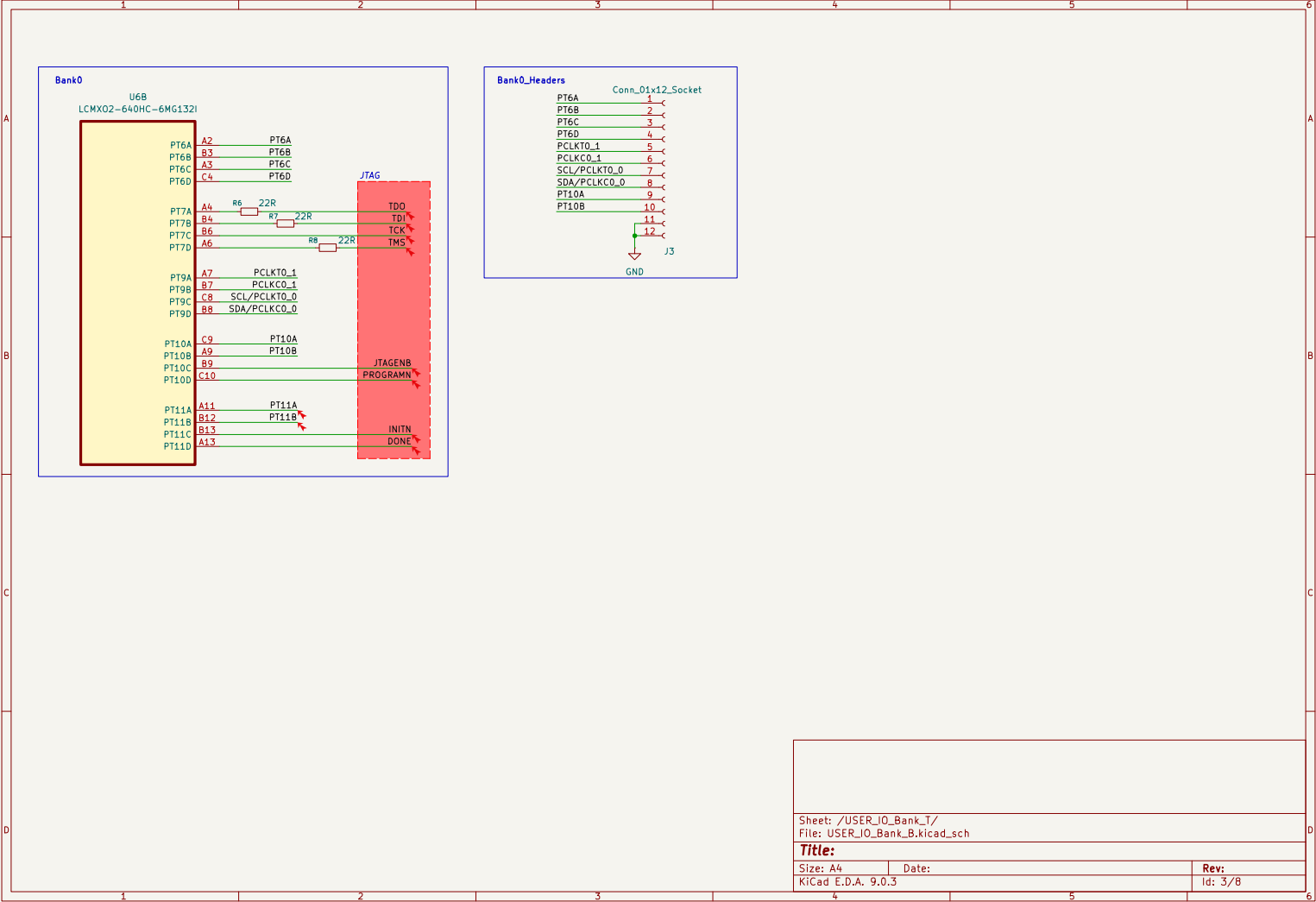
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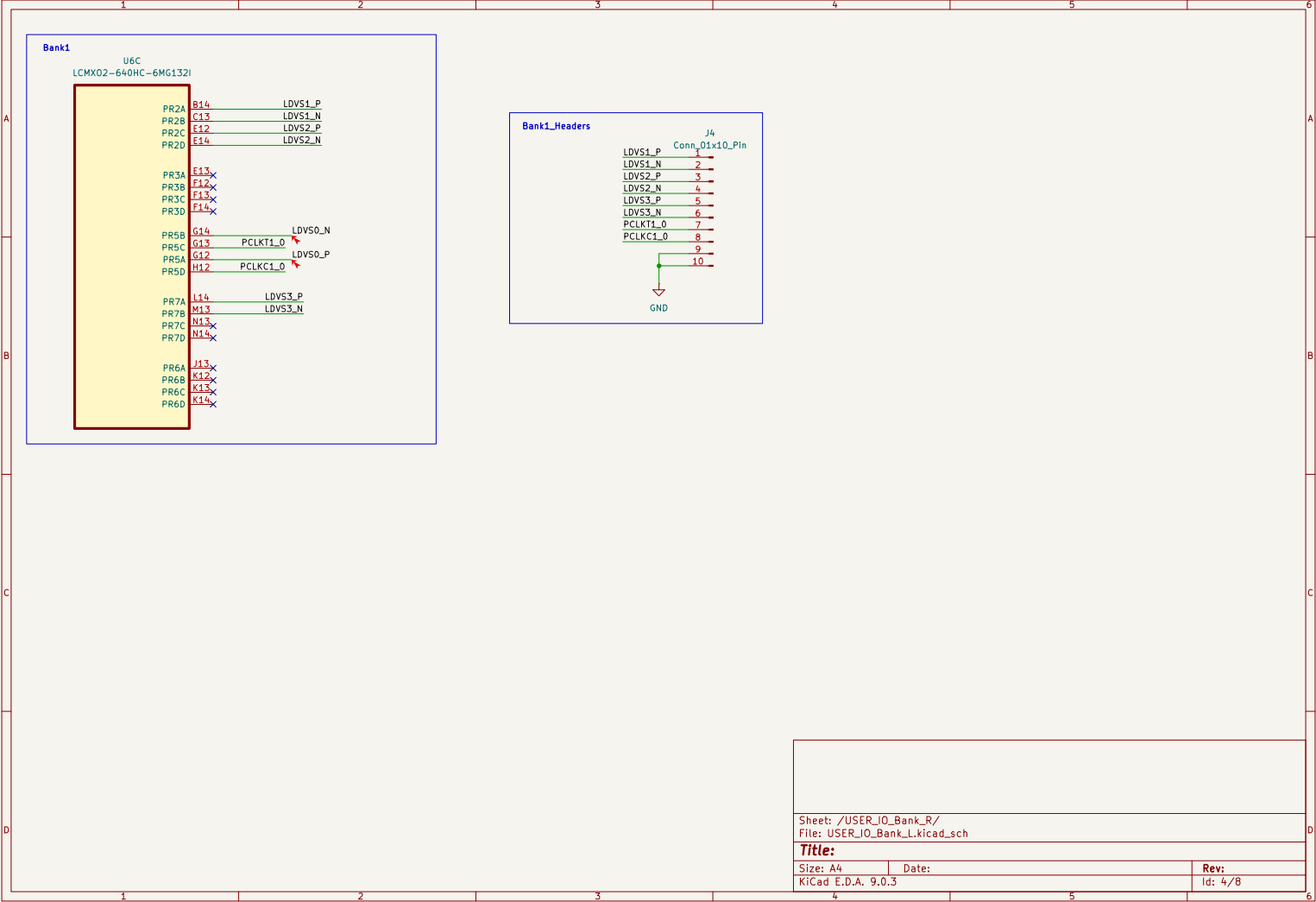
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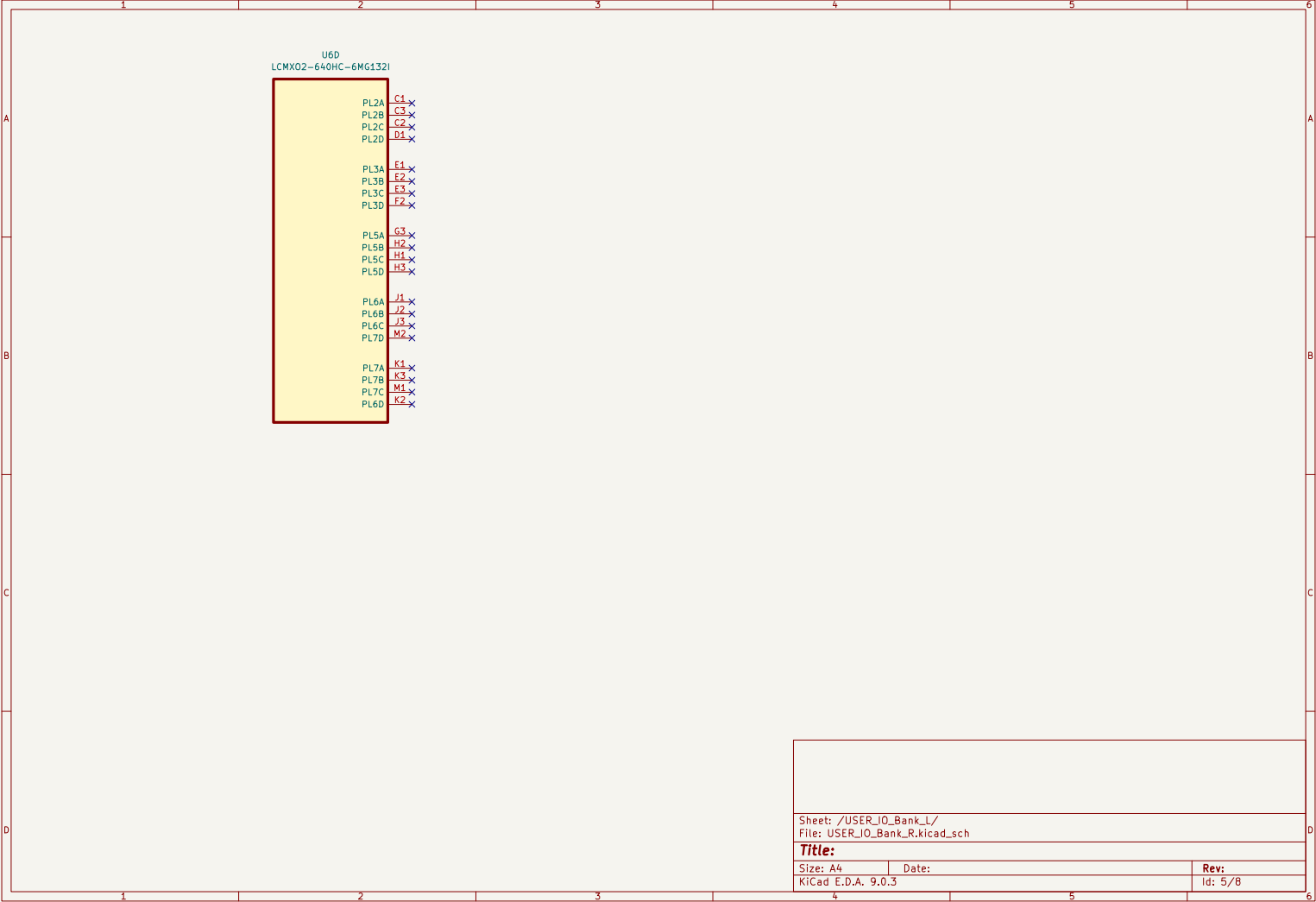
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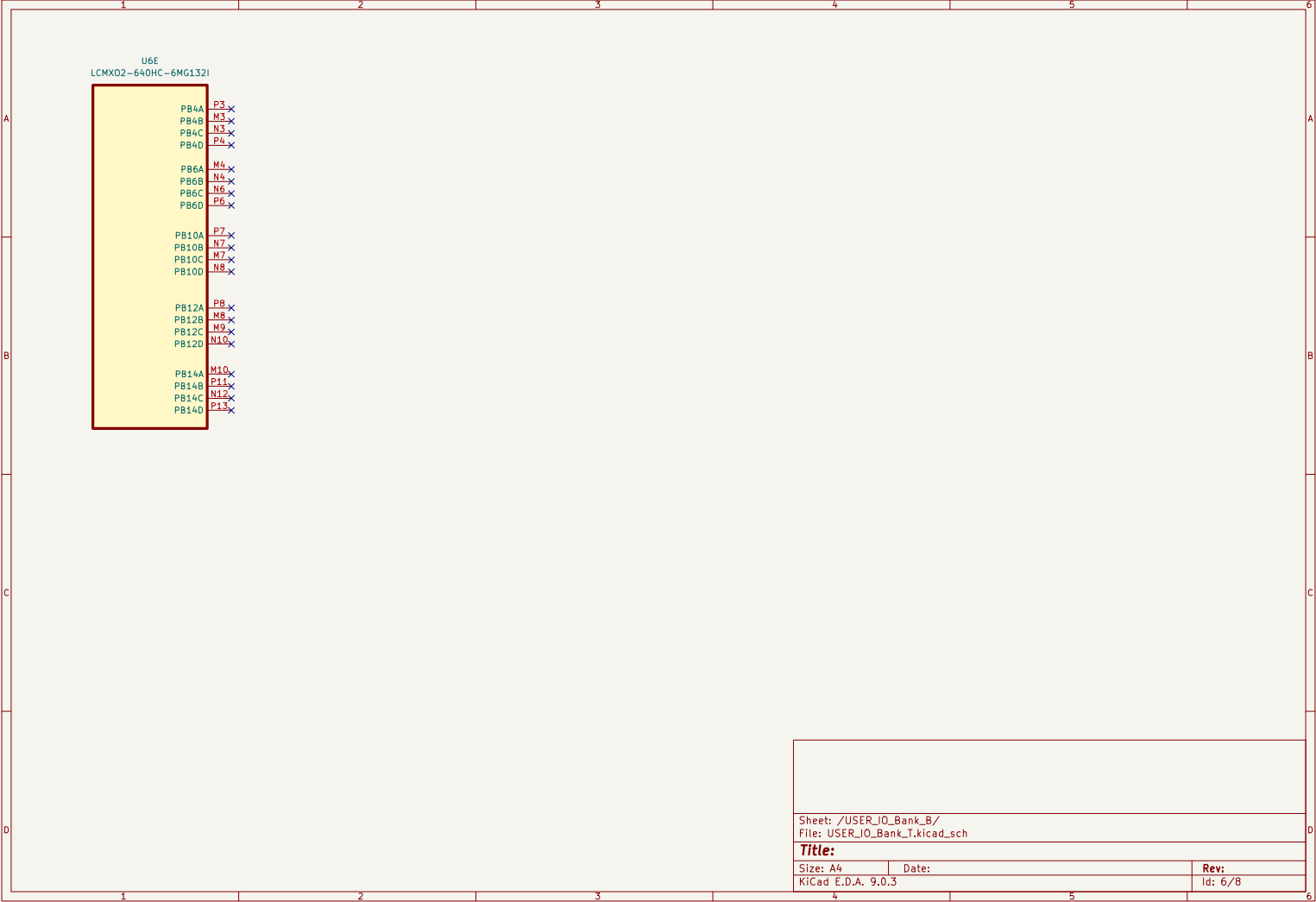
KiCad E.D.A. 9.0.3

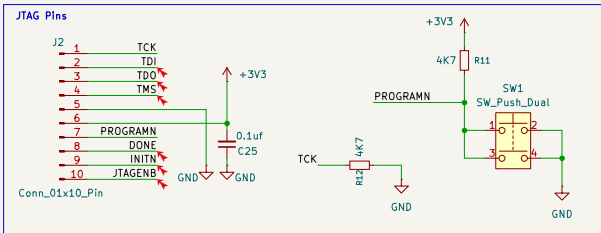
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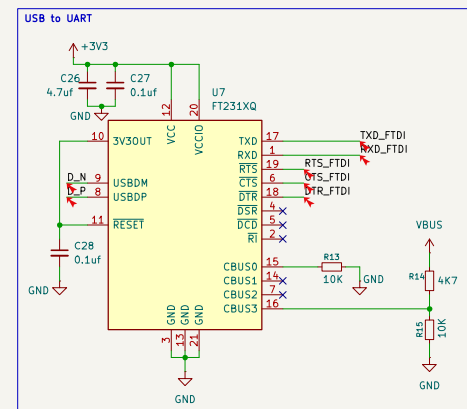


INITN (B13): Open-drain; low during config start – leave floating (internal weak pull-up).
JTAGENB (B9): High = JTAG enabled – leave floating (internal weak pull-up).



Use JTAG Programmer: Lattice HW-USBN-2B USB cable with 10 flywire for programming the FPGA
-It's an official tool for JTAG/SPI programming of MachXO2 FPGAs.

JTAG HEADER (1×10) - LCMXO2-640HC-6MG132I and HW-USBN-2B					
Pin	Signal	FPGA Ball	Flywire	Required?	Notes / Passives
1	TCK	B6	WHITE	YES	4.7 kΩ ↓ to GND
2	TDI	B4	ORANGE	YES	22 Ω series
3	TDO	A4	BROWN	YES	Direct
4	TMS	A6	PURPLE	YES	22 Ω series
5	GND	Any	BLACK	YES	Multi_GND via
6	VCC	VCCIO0	RED	YES	0.1 pF cap
7	PROGRAMN	C10	YELLOW	Opt	4.7 kΩ ↑ + btn
8	DONE	A13	BLUE	Opt	LED + 330 Ω
9	INITN	B13	GREEN	Opt	Float OK
10	JTAGENB	B9	GRAY	Opt	Float OK



Sheet: /USER_Config/
File: USER_Config.kicad_sch

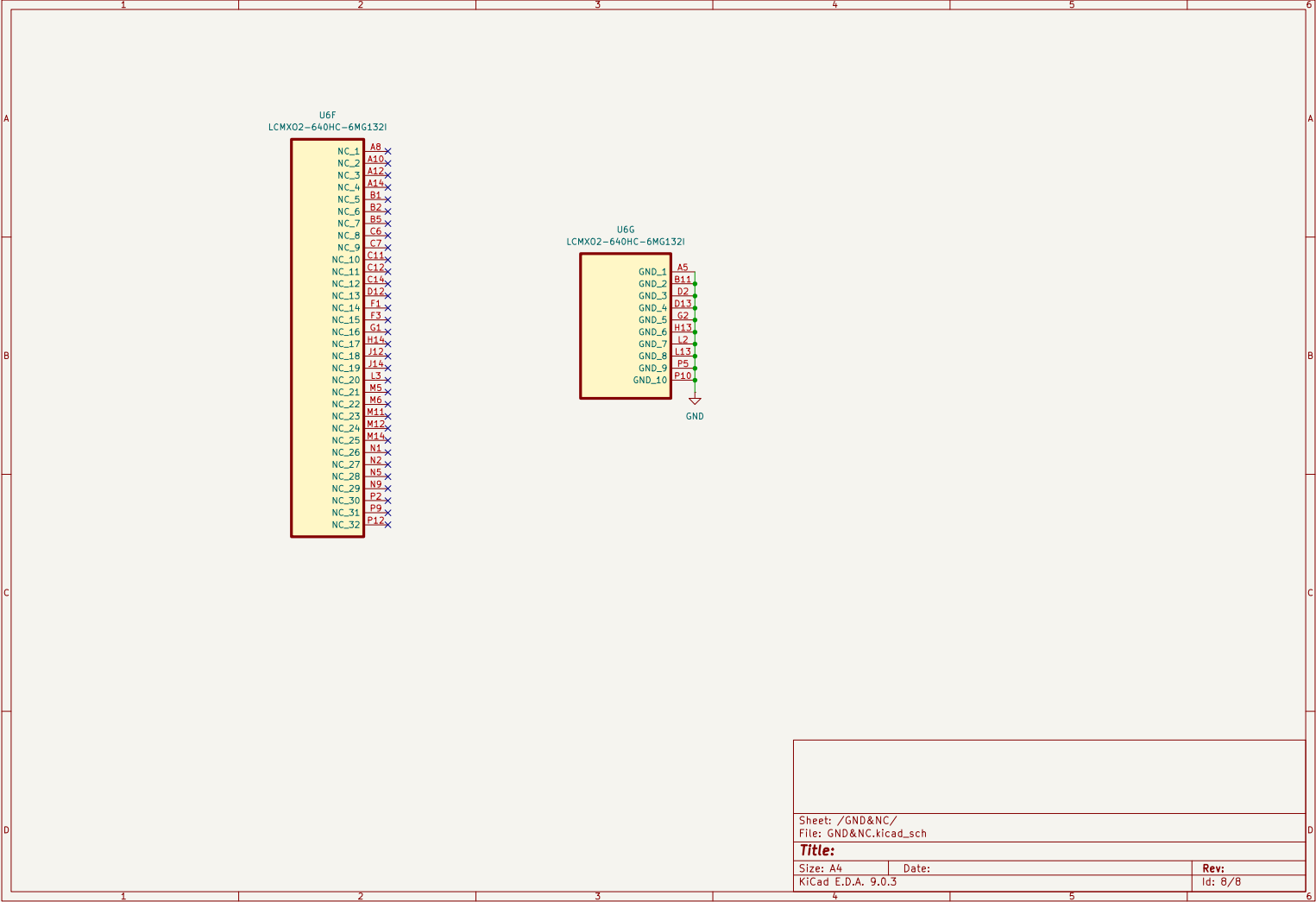
Title:

Size: A4
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Date:

Rev:

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Sheet: /GND&NC/		
File: GND&NC.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.3		Id: 8/8